BRIDGE TO BANGLADESH PLATFORM

Non-Functional Requirements Specification (NFRS)

System Quality, Security, Compliance, Performance & Infrastructure Requirements

Prepared By	Revinr IT Ltd.
Version	v1.3
Date	21 April 2026

Document Version Control

Version	Date	Changes Made	Approved By
V1.1	9th April	Inception Report v1.1	
V1.2	15th April	Revised Inception Report v1.2- Changes discussed during the Oxfam meeting on 9th April, with Phase 1 focused exclusively on the website and diaspora members.	
V1.3	21st April	Separate template as per Oxfam's Instruction	

Table of Contents

Document Version Control 1
1. Introduction and Document Purpose 3
1.1 Purpose 3
1.2 Relationship with FRS and Other Documents 3
1.3 NFRS Quality Dimensions Covered 3
2. Security Requirements 4
3. Data Privacy & GDPR Requirements 7
4. Performance Requirements 10
5. Accessibility Requirements 13
6. Scalability Requirements 15
7. Availability & Recovery Requirements 16
8. Maintainability Requirements 18
9. Internationalisation Requirements 19
10. Hosting & Infrastructure Requirements 20
11. Compliance and Standards Summary 22

1. Introduction and Document Purpose

1.1 Purpose

This Non-Functional Requirements Specification (NFRS) defines how the Bridge to Bangladesh (B2B) platform must perform, not merely what it must do (which is covered by the companion FRS document, B2B-FRS-v1.0). It establishes measurable quality standards, compliance obligations, infrastructure requirements, and performance thresholds that govern the design, development, testing, and long-term operation of the platform.

Every requirement in this document is testable, measurable, and traceable to a source: the Terms of Reference, consultation sessions, regulatory obligations, or architectural design decisions agreed with Oxfam Bangladesh.

1.2 Relationship with FRS and Other Documents

- B2B-FRS-v1.0 — Functional Requirements Specification: defines what the platform must do
- B2B-NFRS-v1.0 (this document): defines how the platform must perform across all quality dimensions
- Inception Report v3.0 — Section 3.4: NFRS scope originally defined here
- Revised Inception Report v1.2 — Chapters 3.2, 4.1, 5: architecture, security, and compliance baseline
- ToR Sections 3.1, 3.7, 4: hosting, security, accessibility, and performance obligations

1.3 NFRS Quality Dimensions Covered

#	Quality Dimension	Summary Scope
1	Security	OWASP Top 10 compliance, encryption, authentication, access control, penetration testing
2	Data Privacy & GDPR	UK GDPR, consent management, data minimisation, right to erasure, portability, cookie compliance
3	Performance	Page load targets, API response thresholds, concurrent user capacity, CDN, caching strategy
4	Accessibility	WCAG 2.1 Level AA, screen reader compatibility, bilingual support, cognitive accessibility
5	Scalability	Horizontal scaling, microservices independence, auto-scaling, future growth headroom
6	Availability & Recovery	Uptime SLA, RTO, RPO, disaster recovery, automated failover, backup strategy
7	Maintainability	Modular codebase, code documentation, CI/CD pipeline, admin interface usability
8	Internationalisation	English and Bengali language support, Unicode, timezone handling, low-bandwidth optimisation
9	Hosting & Infrastructure	Cloud platform (AWS/Azure), environment strategy, free 2-year hosting commitment, monitoring

2. Security Requirements

Security is a foundational design principle for the Bridge to Bangladesh platform, not a layer added on top. The platform adopts a defence-in-depth approach: security controls are enforced simultaneously at the network, infrastructure, API, application, and data levels. All security requirements below are non-negotiable and must be met before the platform is promoted to production.

Requirement ID	NFRS-SEC-001
Quality Area	**Security**
Title	OWASP Top 10 Compliance
Requirement Statement	The platform must be fully compliant with the OWASP Top 10 web application security risks at the point of production go-live and throughout the maintenance period.
Rationale	OWASP Top 10 represents the most critical web application security risks. Full compliance is required by ToR and is non-negotiable given the platform handles personal data of vulnerable populations across multiple jurisdictions.
Measurable Target / Threshold	Zero OWASP Top 10 critical or high vulnerabilities at go-live. Verified by independent penetration test. All medium-severity findings resolved or formally risk-accepted by Oxfam before go-live. Automated OWASP scanning integrated in CI/CD pipeline — build fails on new critical findings.
Test / Verification Method	Independent third-party penetration test covering OWASP Top 10 scope. Automated DAST (Dynamic Application Security Testing) via OWASP ZAP in CI/CD pipeline. Manual security review by QA & Security Tester at end of each sprint.
Risk if Not Met	Platform security breach; personal data of diaspora users and local actors exposed; regulatory penalties under UK GDPR; reputational damage to Oxfam.
Source	ToR Section 3; Inception Report v3.0 Section 3.4; Revised Inception Report v1.2 Section 5.2; GDPR/Security Alignment Session

Requirement ID	NFRS-SEC-002
Quality Area	**Security**
Title	Encryption — Data at Rest and in Transit
Requirement Statement	All data stored on the platform must be encrypted at rest. All data transmitted between clients and servers, and between internal services, must be encrypted in transit.
Rationale	Encryption at rest protects against data breaches from storage compromise. Encryption in transit protects against interception. Both are required for GDPR compliance and Oxfam information security policy.
Measurable Target / Threshold	At rest: AES-256 encryption for all PostgreSQL databases (AWS RDS encryption) and MongoDB Atlas. At transit: TLS 1.3 enforced for all external connections. gRPC with TLS for all internal service-to-service communication. HTTPS enforced — HTTP connections are automatically redirected. No sensitive data (tokens, passwords, PII) stored in URLs or server logs.
Test / Verification Method	Infrastructure configuration audit. TLS configuration scan using sslyze or equivalent. Database encryption configuration review. Penetration test includes encrypted connection verification.
Risk if Not Met	Data breach risk elevated. GDPR compliance failure — encryption is explicitly referenced as a technical measure in UK GDPR Article 32.

Source	UK GDPR Article 32; GDPR/Security Alignment Session; Revised Inception Report v1.2 Section 5.2

Requirement ID	NFRS-SEC-003
Quality Area	**Security**
Title	Multi-Factor Authentication (MFA) Enforcement
Requirement Statement	MFA must be mandatory for all Admin, Moderator, and Auditor roles. It must be offered as a strongly encouraged opt-in for all other user roles.
Rationale	Admin and Moderator accounts have elevated privileges that could cause significant platform harm if compromised. MFA is the most effective control against credential-based attacks and is required by Oxfam's information security policy.
Measurable Target / Threshold	MFA is enforced at login for Admin, Moderator, and Auditor roles — login cannot be completed without MFA verification code. MFA method: TOTP (Time-based One-Time Password) via authenticator app. Account lockout: 5 failed login attempts triggers a 15-minute lockout with admin notification. Backup codes: 10 backup codes generated at MFA setup, invalidated on use.
Test / Verification Method	Security test: attempt Admin login without MFA — system must reject. Penetration test includes brute force and credential stuffing scenarios. Review of lockout logs during QA.
Risk if Not Met	Compromised admin accounts could result in mass data breach, platform misuse, or content sabotage.
Source	ToR Section 3; GDPR/Security Alignment Session; Inception Report v3.0 Section 3.4

Requirement ID	NFRS-SEC-004
Quality Area	**Security**
Title	OWASP Top 10 — Specific Mitigations
Requirement Statement	Each of the OWASP Top 10 risk categories must have a documented and implemented technical mitigation applied to the platform.
Rationale	Compliance with the OWASP Top 10 requires not just testing but implementation of specific controls. This requirement documents the controls that must be in place.
Measurable Target / Threshold	A01 Broken Access Control: RBAC at API Gateway; all endpoints require authenticated JWT; admin behind 2FA-protected login. A02 Cryptographic Failures: AES-256 at rest; TLS 1.3 in transit; bcrypt for password hashing. A03 Injection: Prisma ORM throughout — no raw SQL; parameterised queries mandatory; input validation on all entry points. A04 Insecure Design: Threat modelling at architecture phase; security requirements defined before feature development. A05 Security Misconfiguration: Infrastructure-as-code with version control; secrets in secrets manager, never in source code. A06 Vulnerable Components: Automated dependency scanning in CI/CD; no end-of-life libraries. A07 Authentication Failures: MFA mandatory; JWT expiry and rotation; account lockout. A08 Data Integrity Failures: Signed deployment artefacts; immutable audit logs. A09 Logging and Monitoring: Centralised logging (ELK Stack); real-time alerting on anomalous activity. A10 SSRF: Outbound request allowlisting; no user-controlled URL fetching; external integrations via API Gateway only.
Test / Verification Method	Independent penetration test against all 10 categories. CI/CD pipeline OWASP ZAP scan. Code review checklist aligned to OWASP controls. Pre-go-live security sign-off report.

Risk if Not Met	Any unmitigated OWASP risk creates exploitable vulnerability in production.
Source	Revised Inception Report v1.2 Section 5.2; GDPR/Security Alignment Session

Requirement ID	NFRS-SEC-005
Quality Area	**Security**
Title	Immutable Audit Trail
Requirement Statement	All administrative actions, moderation decisions, financial transactions, and user consent events must be recorded in an immutable audit trail.
Rationale	An immutable audit trail is required for regulatory compliance (GDPR accountability principle), financial governance (donation records), and platform safeguarding (moderation decisions must be reviewable). It is also required for Oxfam internal governance and donor reporting.
Measurable Target / Threshold	Audit log records: action_type, performed_by (user_id), target_entity, target_id, ip_address, timestamp, outcome. All records are append-only — no modification or deletion is permitted. Audit logs are stored separately from operational data. Logs are retained for minimum 7 years (financial actions) and 3 years (all other actions). Logs are indexed in ELK Stack for search and review. Admin can view and export audit logs. No admin can edit or delete audit log entries — including superadmin.
Test / Verification Method	Audit log integrity test: attempt to modify audit log record via API — system must reject. Review of log schema and permissions. Penetration test includes audit log tampering attempt.
Risk if Not Met	Loss of accountability; inability to investigate complaints or security incidents; GDPR compliance failure (accountability principle Article 5(2)).
Source	UK GDPR Article 5(2); ToR Section 3.10; Donation/Financial Requirements Session; Oxfam Key Management Meeting 19 Feb 2026

3. Data Privacy & GDPR Requirements

The Bridge to Bangladesh platform serves users across the United Kingdom, European Union, North America, and Asia-Pacific. Data protection compliance is governed primarily by UK GDPR (UK Data Protection Act 2018) and, where applicable, EU GDPR for EU-resident users. These requirements were confirmed as non-negotiable by Mr Ashish Dammle (Oxfam Bangladesh GDPR focal point) in the GDPR/Security Alignment Session.

Requirement ID	NFRS-GDPR-001
Quality Area	**Data Privacy & GDPR**
Title	Data Minimisation
Requirement Statement	The platform must collect only data that is essential to the platform's stated functions. No speculative, pre-emptive, or unnecessary data collection is permitted.
Rationale	UK GDPR Article 5(1)(c) — data minimisation principle. Required as a foundational design principle, not an afterthought. Platform serves vulnerable populations in some cases; data minimisation reduces risk of harm if a breach occurs.
Measurable Target / Threshold	Profile fields are structured so the minimum viable dataset is captured at registration, with optional progressive completion thereafter. No speculative data fields are permitted in any schema. All data fields must be justified in a data inventory document maintained by Revinr IT. Data inventory to be reviewed and approved by Oxfam GDPR focal point before go-live.
Test / Verification Method	Review of registration and profile forms against data inventory. Code review of all schema definitions. DPA (Data Protection Assessment) conducted before go-live.
Risk if Not Met	GDPR enforcement action; reputational damage; loss of user trust.
Source	UK GDPR Article 5(1)(c); GDPR/Security Alignment Session; Revised Inception Report v1.2 Section 5.3.1

Requirement ID	NFRS-GDPR-002
Quality Area	**Data Privacy & GDPR**
Title	Lawful Basis and Granular Consent Management
Requirement Statement	The platform must implement granular, specific consent management for all data processing activities, with consent events recorded in an auditable consent ledger.
Rationale	UK GDPR requires a clearly identified lawful basis for each data processing activity. For the B2B platform, the primary lawful basis is consent for non-essential processing. Consent must be specific, informed, freely given, and withdrawable at any time.
Measurable Target / Threshold	Consent captured at registration for: platform use (service delivery), GDPR/data handling, marketing communications. Each consent type is stored independently in the consents table with: consent_type, accepted_at timestamp, scope, version of privacy policy accepted. Granular controls in account settings: users can update or withdraw any individual consent without withdrawing all. Consent cannot be bundled — each type must be separately opted into. Consent withdrawal is actionable by the user without contacting Oxfam staff.
Test / Verification Method	Review of consents table schema. Test: withdraw marketing consent only — platform access must be unaffected. Test: attempt to send marketing communication to user who has withdrawn marketing consent — system must block. Consent audit log review.

Risk if Not Met	GDPR enforcement; ICO investigation; loss of platform trust; inability to demonstrate lawful basis to regulators.
Source	UK GDPR Articles 6, 7, 13; GDPR/Security Alignment Session; Revised Inception Report v1.2 Section 5.3.2

Requirement ID	NFRS-GDPR-003
Quality Area	**Data Privacy & GDPR**
Title	Right to Erasure (Right to be Forgotten)
Requirement Statement	The platform must implement a complete and verifiable right to erasure workflow, enabling users to request deletion of all their personal data.
Rationale	UK GDPR Article 17 — right to erasure. Users have a legal right to request deletion of their personal data. The platform must be able to action this request completely, including across all microservices that hold data relating to the user.
Measurable Target / Threshold	Right to erasure request is available from account settings without contacting Oxfam staff. On submission: user account is deactivated immediately; all PII fields are replaced with null/anonymised values within 30 days; cascade deletion procedures run across all microservices (profiles, connections, messages, content, consent records, donations — donor records anonymised but financial totals retained for audit); erasure event is recorded in audit log; user receives confirmation email. Donation records: anonymised (PII removed) but transaction amounts and fund references retained for financial audit. Content published by the user: authorship anonymised; content remains if removed content would break platform context — or removed entirely at Oxfam discretion.
Test / Verification Method	End-to-end erasure test: create test user with data across all modules, submit erasure request, verify all PII removed within 30 days. Audit log review for erasure event. Penetration test: attempt to access erased user data via API.
Risk if Not Met	GDPR enforcement; ICO investigation; legal liability; reputational damage.
Source	UK GDPR Article 17; GDPR/Security Alignment Session; Revised Inception Report v1.2 Section 5.3.3

Requirement ID	NFRS-GDPR-004
Quality Area	**Data Privacy & GDPR**
Title	Data Portability
Requirement Statement	The platform must enable users to export all personal data held about them in a structured, commonly used, machine-readable format.
Rationale	UK GDPR Article 20 — right to data portability. Users must be able to take their data and transfer it to another service.
Measurable Target / Threshold	Data export function available in account settings. Export includes: profile data, connection history, content authored, campaign participation, consent history, donation history (anonymised where appropriate). Format: JSON (primary) and CSV (summary). Export is generated asynchronously — user receives email with secure download link within 24 hours. Download link expires after 48 hours. Export request is logged in audit trail.
Test / Verification Method	Test: request data export — verify JSON/CSV file contains all expected data fields. Test: download link expires after 48 hours. Test: only the requesting user can access their export link.

Risk if Not Met	GDPR enforcement; loss of user trust; legal liability.
Source	UK GDPR Article 20; GDPR/Security Alignment Session; Revised Inception Report v1.2 Section 5.3.4

Requirement ID	NFRS-GDPR-005
Quality Area	**Data Privacy & GDPR**
Title	Cookie Compliance
Requirement Statement	The platform must implement a compliant cookie consent mechanism that meets UK GDPR and PECR requirements.
Rationale	UK PECR (Privacy and Electronic Communications Regulations) requires informed consent before setting non-essential cookies. This applies to the web application.
Measurable Target / Threshold	Cookie consent banner displayed on first visit. Categories: strictly necessary (no consent required), analytics, marketing. Users can accept all, reject all, or customise by category. Cookie preferences are stored per user (authenticated) or per device (unauthenticated). Preferences are respected immediately — no analytics or marketing cookies set before consent. Cookie audit: full inventory of all cookies used on the platform, categorised correctly, reviewed before go-live.
Test / Verification Method	Cookie audit review. Test: inspect browser cookies before and after consent withdrawal — non-essential cookies must not be present before consent. Third-party cookie audit.
Risk if Not Met	ICO enforcement; reputational damage; loss of user trust.
Source	UK PECR; UK GDPR; GDPR/Security Alignment Session; Revised Inception Report v1.2 Section 5.3.6

4. Performance Requirements

Performance targets have been defined based on the platform's global user base — diaspora members across 12+ countries and Bangladesh-based local actors. Bandwidth conditions vary significantly, particularly for users in Bangladesh. Targets are set conservatively to ensure a reliable experience across all user contexts.

Requirement ID	NFRS-PERF-001
Quality Area	**Performance**
Title	Page Load Time
Requirement Statement	All primary pages of the web application must load within 3 seconds on a standard 4G mobile connection under normal load conditions.
Rationale	Diaspora users span multiple geographies and network conditions. Bangladesh-based users may have lower bandwidth. A 3-second load threshold is the established standard for acceptable web application performance and directly impacts user retention.
Measurable Target / Threshold	Page load time ≤ 3 seconds (Time to Interactive) on a simulated 4G connection (12 Mbps download) for: Home, Directory/Search, Profile Detail, Campaign Pages, Content Hub. Measured using Google Lighthouse (target score: Performance ≥ 85). Core Web Vitals targets: LCP (Largest Contentful Paint) ≤ 2.5 seconds; FID (First Input Delay) ≤ 100ms; CLS (Cumulative Layout Shift) ≤ 0.1.
Test / Verification Method	Google Lighthouse automated testing in CI/CD pipeline. Manual performance testing pre-launch using WebPageTest with simulated 4G conditions. Load test with 500 concurrent simulated users.
Risk if Not Met	Poor performance drives diaspora user abandonment. Platform perceived as unreliable; engagement targets not met.
Source	Revised Inception Report v1.2 Section 3.2 — Performance; Consultation Notes UK Diaspora

Requirement ID	NFRS-PERF-002
Quality Area	**Performance**
Title	API Response Time
Requirement Statement	All standard API endpoints must respond within 500 milliseconds under normal load. Search and matchmaking endpoints must respond within 1 second.
Rationale	API response time directly determines the perceived responsiveness of both the web and mobile applications. Sub-500ms responses are industry standard for a responsive experience.
Measurable Target / Threshold	Standard API endpoints (profile retrieval, content listing, campaign listing): P95 response time ≤ 500ms under normal load (up to 500 concurrent users). Search endpoints: P95 ≤ 1000ms. Matchmaking recommendation endpoint: P95 ≤ 1500ms (score retrieval from Redis cache). Payment gateway calls: P95 ≤ 3000ms (dependent on gateway SLA). Caching strategy: Redis caching applied to all high-frequency, low-change read operations (profile lookups, match scores, SDG taxonomy).
Test / Verification Method	API load testing using Apache JMeter or k6 with 500 concurrent virtual users. Performance profiling per endpoint in development and staging. Redis cache hit rate monitoring (target: >80% hit rate for cached endpoints).

Risk if Not Met	Slow API responses degrade mobile app experience; increases user frustration; may cause timeout errors.
Source	Revised Inception Report v1.2 Section 3.2 — Performance

Requirement ID	NFRS-PERF-003
Quality Area	**Performance**
Title	Concurrent User Capacity
Requirement Statement	The platform must support a minimum of 500 concurrent active users at launch without performance degradation, with architecture capable of scaling to 5,000 concurrent users.
Rationale	Platform targets diaspora communities across 12+ countries in multiple time zones. Concurrent peak usage must be planned for. Phase 1 initial user base estimate: 2,000–5,000 registered users; 500 concurrently active at peak.
Measurable Target / Threshold	Load test scenario: 500 concurrent virtual users performing realistic actions (browsing, searching, viewing profiles) simultaneously. API response times must remain within NFRS-PERF-002 thresholds at 500 concurrent users. Auto-scaling configuration: cloud infrastructure auto-scales beyond 500 concurrent users without manual intervention. Horizontal scaling via cloud load balancer and container orchestration.
Test / Verification Method	Load testing using k6 or Apache JMeter: 500 concurrent virtual users, 30-minute sustained load test. Monitoring: Prometheus/Grafana dashboards reviewed during and after load test. Auto-scaling test: simulate traffic spike to 1000 concurrent users and verify auto-scaling responds within 2 minutes.
Risk if Not Met	Platform unresponsive or slow during high-profile campaigns or events; reputational damage; diaspora user abandonment.
Source	Technical Proposal — Revinr IT Ltd; Inception Report v3.0

Requirement ID	NFRS-PERF-004
Quality Area	**Performance**
Title	Low-Bandwidth Optimisation
Requirement Statement	The platform must be optimised for users on lower-bandwidth connections, particularly Bangladesh-based local actors and diaspora users in regions with variable connectivity.
Rationale	Bangladesh connectivity can be significantly lower than UK/EU averages. Local actors are a primary user group and must not be disadvantaged by infrastructure limitations. This was raised explicitly in consultations.
Measurable Target / Threshold	Images served via CDN (AWS CloudFront / Azure CDN) with automatic format optimisation (WebP where supported). Images lazy-loaded; above-fold content prioritised. Video content: progressive streaming; no autoplay on mobile. Mobile app (Flutter): offline-capable core features (profile view, saved profiles); synchronised when connectivity restored. CDN coverage: edge nodes covering UK, EU, North America, and Asia-Pacific. First Contentful Paint target: ≤ 1.5 seconds on 3G connection for home page.
Test / Verification Method	Testing on throttled connections (3G simulation in Chrome DevTools). CDN configuration review. Core Web Vitals measurement on slow connection simulation.

Risk if Not Met	Local actors in Bangladesh experience slow platform; engagement with diaspora connections reduced; platform perceived as unusable.
Source	Consultation Notes UK Diaspora; Local NGO/CSO Workshop 12 March 2026; Revised Inception Report v1.2

5. Accessibility Requirements

Accessibility is mandated by the Terms of Reference at WCAG 2.1 Level AA. The diaspora user demographic is broad in age range and digital literacy. Local actors in Bangladesh may use a variety of devices and screen sizes. Accessibility is not optional — it is a contractual requirement.

Requirement ID	NFRS-ACC-001
Quality Area	**Accessibility**
Title	WCAG 2.1 Level AA Compliance
Requirement Statement	The web application must fully comply with WCAG 2.1 Level AA accessibility standards across all core interface elements.
Rationale	WCAG 2.1 AA is mandated by ToR Section 3.2. Diaspora user base spans a broad age range. Users with visual, motor, or cognitive impairments must be able to use the platform fully. WCAG 2.1 AA is the internationally recognised standard for accessible web applications.
Measurable Target / Threshold	All pages pass automated WCAG 2.1 AA scanning (zero critical or serious violations). Colour contrast ratio: minimum 4.5:1 for normal text; 3:1 for large text. All interactive elements are keyboard-navigable. All images have descriptive alt text. All form inputs have associated labels. All error messages are descriptive and associated with the relevant form field. Focus indicators are clearly visible.
Test / Verification Method	Automated accessibility scanning using axe-core or Lighthouse in CI/CD pipeline — build fails on WCAG 2.1 AA critical violations. Manual screen reader testing (NVDA + Chrome, VoiceOver + Safari). Keyboard-only navigation test for all core user journeys. Pre-launch third-party accessibility audit.
Risk if Not Met	Users with disabilities unable to access platform; potential Equality Act (UK) liability; reputational damage; ToR breach.
Source	ToR Section 3.2; Inception Report v3.0 Section 3.4; Revised Inception Report v1.2 Section 3.2

Requirement ID	NFRS-ACC-002
Quality Area	**Accessibility**
Title	Screen Reader Compatibility
Requirement Statement	All core interface elements must be compatible with leading screen readers.
Rationale	Visually impaired users require screen reader compatibility to access the platform independently. This is a WCAG 2.1 AA requirement and a contractual obligation.
Measurable Target / Threshold	Screen reader compatibility verified for: NVDA + Chrome (Windows), VoiceOver + Safari (macOS/iOS), TalkBack (Android). All interactive elements have ARIA labels where semantic HTML is insufficient. Logical reading order maintained throughout all pages. Dynamic content updates (notifications, search results) announce correctly to screen readers via ARIA live regions.
Test / Verification Method	Manual screen reader testing with NVDA + Chrome and VoiceOver + Safari for all core user journeys: registration, login, search, profile view, connection request, content reading, campaign pledge.
Risk if Not Met	Visually impaired users excluded from platform; ToR non-compliance.
Source	ToR Section 3.2; WCAG 2.1 AA Success Criterion 4.1.2

Requirement ID	NFRS-ACC-003
Quality Area	**Accessibility**
Title	Adjustable Text and Contrast
Requirement Statement	The platform must support text size adjustment and provide high-contrast mode for users with visual impairments.
Rationale	Users with low vision or age-related visual decline require the ability to increase text size without losing content or functionality. High contrast mode improves readability for users with visual impairments.
Measurable Target / Threshold	Text resizes up to 200% (using browser zoom) without loss of content or functionality. Adjustable font size controls provided in user profile settings (small / medium / large). High-contrast mode available in user profile settings. No text presented as images (except logos).
Test / Verification Method	Test: increase browser zoom to 200% on all core pages — no horizontal scrolling on desktop; no content loss. Test: enable high-contrast mode — verify all content remains readable. Automated contrast ratio check using axe-core.
Risk if Not Met	Users with visual impairments unable to read content; platform inaccessible to older diaspora users.
Source	WCAG 2.1 AA Success Criteria 1.4.4, 1.4.6; ToR Section 3.2

6. Scalability Requirements

Requirement ID	NFRS-SCALE-001
Quality Area	**Scalability**
Title	Horizontal Scaling via Microservices Architecture
Requirement Statement	The platform architecture must support horizontal scaling of individual services without requiring full platform re-deployment or re-platforming.
Rationale	The microservices architecture is specifically chosen to enable independent scaling of services under differential load. Campaign-driven traffic spikes may cause the Campaign Service to require additional instances while other services remain unaffected. Monolithic architecture cannot achieve this.
Measurable Target / Threshold	Each of the eleven microservices can be scaled independently by increasing container replicas. Cloud auto-scaling is configured per service based on CPU and memory thresholds (default trigger: >70% CPU utilisation for 2 consecutive minutes). PostgreSQL read replicas provisioned for high-traffic read-heavy services (profiles, matchmaking, content). MongoDB Atlas sharding configured and ready to activate. Redis cluster mode available for cache scaling.
Test / Verification Method	Load test with differential service stress (simulate campaign launch spike for Campaign Service only) — verify other services are unaffected. Auto-scaling test: simulate CPU spike on Profile Service — verify auto-scaling responds within 2 minutes. Review of cloud auto-scaling configuration.
Risk if Not Met	Platform unable to handle growth; expensive full re-platforming required to scale; poor performance during high-profile campaign launches.
Source	Revised Inception Report v1.2 — Architecture Section 4.1; Technical Proposal — Revinr IT Ltd

Requirement ID	NFRS-SCALE-002
Quality Area	**Scalability**
Title	Future Growth Architecture — Phase 2 Readiness
Requirement Statement	The platform architecture must be designed to accommodate Phase 2 features (additional modules, increased user base, new geographies) without fundamental architectural changes.
Rationale	The ToR and Inception Report make clear that Phase 1 is the first phase of a longer-term platform. The architecture must not create technical debt that blocks Phase 2 development.
Measurable Target / Threshold	New microservices can be added to the platform following the established service pattern (own data store, API Gateway registration, RBAC integration) without modifying existing services. Database schema uses UUID primary keys throughout (no auto-increment integers) to support future distributed architectures. API versioning (v1, v2 prefix) is implemented from launch. Feature flags architecture supports progressive rollout of Phase 2 features without full deployment.
Test / Verification Method	Architecture review at Milestone 4 handover — documentation must demonstrate Phase 2 addition patterns. API versioning reviewed in code. UUID implementation verified across all schemas.
Risk if Not Met	Phase 2 development blocked by technical debt; expensive rearchitecting required; Oxfam unable to evolve platform sustainably.
Source	ToR Section 3; Inception Report v3.0 — Strategic Direction

7. Availability & Recovery Requirements

Requirement ID	NFRS-AVAIL-001
Quality Area	**Availability & Recovery**
Title	Uptime SLA
Requirement Statement	The production platform must maintain a minimum uptime of 99.5% monthly, excluding planned maintenance windows.
Rationale	The platform serves diaspora users globally across multiple time zones (UK, USA, Australia, Middle East). Significant downtime would erode user trust and disrupt engagement. 99.5% uptime = maximum ~3.6 hours unplanned downtime per month.
Measurable Target / Threshold	Monthly uptime ≥ 99.5% (measured by third-party uptime monitoring). Planned maintenance windows: communicated to users and Oxfam admin at least 48 hours in advance; scheduled during lowest-traffic periods (02:00–04:00 UTC Sunday). Uptime monitoring: UptimeRobot or Pingdom (third-party, independent monitoring) with real-time alerts. Revinr IT responsible for uptime SLA for 2 years from production go-live per ToR Section 3.7.
Test / Verification Method	Monthly uptime reports generated from monitoring dashboard. Incident log reviewed monthly. SLA breach triggers root cause analysis report to Oxfam within 5 business days.
Risk if Not Met	Extended downtime damages diaspora user trust; campaign launches disrupted; Oxfam reporting impacted; ToR SLA breach.
Source	ToR Section 3.7; Technical Proposal — Revinr IT Ltd; Revised Inception Report v1.2

Requirement ID	NFRS-AVAIL-002
Quality Area	**Availability & Recovery**
Title	Recovery Time Objective (RTO) and Recovery Point Objective (RPO)
Requirement Statement	The platform must meet defined RTO and RPO targets to ensure rapid recovery from system failures.
Rationale	Formal RTO and RPO targets are required for disaster recovery planning. They define the maximum acceptable downtime (RTO) and maximum acceptable data loss (RPO) in a failure scenario.
Measurable Target / Threshold	RTO (Recovery Time Objective): ≤ 4 hours for full platform recovery following a complete infrastructure failure. RPO (Recovery Point Objective): ≤ 1 hour of data loss in a catastrophic failure scenario. Achieved via: automated daily PostgreSQL backups with point-in-time recovery (PITR) capability; MongoDB Atlas continuous backup; backup storage in a separate cloud region; documented restoration procedures tested annually.
Test / Verification Method	Disaster recovery drill: simulate database failure and verify restoration within RTO target. Review of backup schedule and retention policy. PITR test: restore database to a point 30 minutes in the past.
Risk if Not Met	Significant data loss or extended downtime in a failure event; reputational damage; loss of user trust; financial reporting gaps.
Source	ToR Section 3.7; Technical Proposal — Revinr IT Ltd; Revised Inception Report v1.2 Section 4.1.8

Requirement ID	NFRS-AVAIL-003
Quality Area	**Availability & Recovery**
Title	Automated Daily Backups
Requirement Statement	All databases must be backed up automatically on a daily basis, with backups stored in a separate region and retained for a minimum defined period.
Rationale	Daily automated backups are the minimum safeguard against data loss due to accidental deletion, corruption, or security incidents. Separate-region storage protects against regional infrastructure failures.
Measurable Target / Threshold	PostgreSQL: automated daily backups via AWS RDS automated backup or Azure Database backup. PITR enabled (minimum 7-day retention window for PITR). Backup retention: 30 days for daily snapshots; 365 days for monthly snapshots. MongoDB Atlas: continuous backup with PITR. Backup storage: separate AWS region or Azure region from primary. Backup encryption: same standards as primary data (AES-256 at rest). Backup restoration tested quarterly.
Test / Verification Method	Quarterly backup restoration test: restore a backup to a test environment and verify data integrity. Review of backup schedule, retention policy, and storage location configuration.
Risk if Not Met	Irrecoverable data loss in a failure scenario; regulatory issues if financial or personal data is lost.
Source	ToR Section 3.7; Technical Proposal — Revinr IT Ltd

8. Maintainability Requirements

Requirement ID	NFRS-MAINT-001
Quality Area	**Maintainability**
Title	Modular, Documented Codebase
Requirement Statement	The platform codebase must be modular, consistently structured, and fully documented to enable handover to Oxfam or a successor vendor.
Rationale	The ToR requires handover at Milestone 4. The codebase must be maintainable by Oxfam technical partners or a successor vendor after Revinr's engagement ends. A poorly documented codebase creates long-term dependency and cost.
Measurable Target / Threshold	All microservices follow consistent directory structure and naming conventions documented in a Developer Guide (M4 deliverable). All functions and API endpoints are documented with inline JSDoc/docstrings. README files in all repository directories. API documentation: full OpenAPI (Swagger) specification for all endpoints (M4 deliverable). Test coverage: minimum 80% unit test coverage for all core business logic functions. Code review is mandatory before merge to main branch — enforced via GitHub branch protection rules.
Test / Verification Method	Code review of repository structure against Developer Guide standard. SonarQube (or equivalent) code quality scan — report provided to Oxfam at M4. Test coverage report: verify ≥ 80% for core business logic. OpenAPI documentation review at M4.
Risk if Not Met	Successor vendor or Oxfam partner unable to maintain or extend the platform; long-term dependency on Revinr; increased maintenance costs.
Source	ToR Section 3.9; Inception Report v3.0 — Milestone 4 Deliverables

Requirement ID	NFRS-MAINT-002
Quality Area	**Maintainability**
Title	Non-Technical Admin Interface
Requirement Statement	The Oxfam admin dashboard must be fully operable by non-technical Oxfam staff without requiring developer involvement for routine platform operations.
Rationale	Oxfam Bangladesh staff operating the platform post-handover are not expected to have technical development skills. All routine operations — content moderation, user management, campaign creation, report export — must be achievable through the admin UI.
Measurable Target / Threshold	All routine administrative tasks achievable through the admin dashboard UI without developer access. Tasks included: content moderation, user verification, user role management, campaign creation and management, analytics report export, moderation queue management. No database or code access required for any routine operational task. Admin dashboard includes contextual help text for all key functions. Oxfam staff training delivered before go-live (20–25 June 2026 per Milestone 4 plan).
Test / Verification Method	Usability testing: Oxfam admin staff complete a set of common tasks in the admin dashboard without developer assistance — target success rate ≥ 90%. Training session evaluation feedback.
Risk if Not Met	Oxfam staff dependent on Revinr for routine operations; increased operational cost; platform not sustainable post-handover.
Source	ToR Section 3.9; SRA Workshop 2; Oxfam Key Management Meeting 19 Feb 2026

9. Internationalisation Requirements

Requirement ID	NFRS-I18N-001
Quality Area	**Internationalisation**
Title	Bilingual Interface — English and Bengali
Requirement Statement	The platform must support full bilingual operation in English and Bengali, with Unicode Bengali support throughout.
Rationale	The platform serves both diaspora users (primarily English-dominant) and Bangladesh-based local actors (Bengali-dominant). Bilingual support is essential for platform usability across both primary user groups. Confirmed by UK Diaspora consultations and Local NGO/CSO Workshop.
Measurable Target / Threshold	All primary interface elements are available in both English and Bengali. Users can switch language from a language toggle accessible on all pages. Language preference is stored in user profile and applied on login. Bengali text: Unicode (UTF-8) encoded throughout; Bengali fonts (Kalpurush or equivalent) loaded via CDN. All CMS content is submittable in English and/or Bengali. All system messages (notifications, error messages, confirmation emails) available in both languages. Admin interface: English primary (confirmed with Oxfam).
Test / Verification Method	Test: switch to Bengali language — verify all primary interface elements render correctly in Bengali. Test: submit content in Bengali — verify correct rendering in content listing and detail views. Bengali font load time test on 3G connection.
Risk if Not Met	Bangladesh-based local actors unable to use platform in their primary language; platform inaccessible to non-English-dominant diaspora users.
Source	ToR Section 3.2; Consultation Notes UK Diaspora; Local NGO/CSO Workshop 12 March 2026; Inception Report v3.0 Section 3.4

Requirement ID	NFRS-I18N-002
Quality Area	**Internationalisation**
Title	Timezone Handling
Requirement Statement	All timestamps stored and processed in UTC. Timestamps displayed to users in their local timezone.
Rationale	The platform serves users across 12+ countries and multiple time zones. Events, campaigns, and message timestamps must be meaningful and accurate regardless of where the user is located.
Measurable Target / Threshold	All timestamps in all databases are stored in UTC. All timestamps displayed in the UI are converted to the user's local timezone using browser timezone detection. UTC offset displayed alongside timestamps for clarity in cross-timezone contexts (e.g., event listings). Campaign start/end dates are displayed with timezone reference. Timezone detection is automatic on first login; users can override in profile settings.
Test / Verification Method	Test: create an event with a specific UTC timestamp; view from accounts simulating UK, USA (EST), and Bangladesh timezones — verify correct local time displayed in each case.
Risk if Not Met	Events and campaign deadlines displayed with incorrect times; user confusion; loss of trust.
Source	Revised Inception Report v1.2 Section 3.2 — Performance (UTC timestamps noted)

10. Hosting & Infrastructure Requirements

Requirement ID	NFRS-INFRA-001
Quality Area	**Hosting & Infrastructure**
Title	Cloud Platform and Multi-Zone Deployment
Requirement Statement	The platform must be hosted on a leading cloud provider (AWS primary, Azure contingency) with multi-zone deployment for high availability.
Rationale	Multi-zone deployment eliminates single points of failure in the infrastructure layer. Required for 99.5% uptime SLA (NFRS-AVAIL-001). Cloud hosting is specified in ToR Section 4.
Measurable Target / Threshold	Cloud infrastructure deployed on AWS (primary choice) or Azure (confirmed with Oxfam before deployment). Multi-zone: services deployed across minimum 2 availability zones within the same region. Auto-failover: if primary zone fails, traffic automatically routes to secondary zone within 60 seconds. Load balancer distributes traffic across healthy service instances. Cloud provider: Revinr confirms with Oxfam Bangladesh PMU before infrastructure provisioning. Free hosting for 2 years from production go-live per ToR Section 3.7.
Test / Verification Method	Infrastructure architecture diagram reviewed at M4 handover. Failover test: simulate primary zone failure — verify traffic routes to secondary zone within 60 seconds. Review of cloud provider confirmation documentation.
Risk if Not Met	Single zone failure causes full platform outage; 99.5% SLA breached; diaspora users unable to access platform.
Source	ToR Sections 3.7 and 4; Technical Proposal — Revinr IT Ltd; Revised Inception Report v1.2 Section 4.1.8

Requirement ID	NFRS-INFRA-002
Quality Area	**Hosting & Infrastructure**
Title	Real-Time Monitoring and Alerting
Requirement Statement	The platform must implement real-time monitoring across all services with automated alerting on threshold breaches.
Rationale	Proactive monitoring enables issues to be identified and resolved before they impact users. Required for maintaining the 99.5% uptime SLA. Prometheus/Grafana monitoring stack is specified in the architecture.
Measurable Target / Threshold	Prometheus metrics collection from all microservices: CPU utilisation, memory usage, API error rates, response times, database connection pool status. Grafana dashboards: real-time visualisation of all key metrics. Alerting: automated alerts sent to Revinr technical team on: API error rate >1% for 5 consecutive minutes; CPU utilisation >80% for 5 consecutive minutes; API response time >2x normal baseline for 10 consecutive minutes; uptime monitoring alert from UptimeRobot/Pingdom. Alert acknowledgement: all critical alerts must be acknowledged within 30 minutes during business hours; within 2 hours out of hours.
Test / Verification Method	Review of Prometheus alert rules configuration. Test: simulate API error rate spike — verify alert triggered within 5 minutes. Review of alert acknowledgement logs from staging environment.
Risk if Not Met	Issues undetected until users report them; delayed incident response; SLA breach.
Source	Revised Inception Report v1.2 Section 4.1.8; Technical Proposal — Revinr IT Ltd

Requirement ID	NFRS-INFRA-003
Quality Area	**Hosting & Infrastructure**
Title	CI/CD Pipeline with Automated Testing
Requirement Statement	All code changes must pass through an automated CI/CD pipeline before reaching the production environment.
Rationale	A fully automated CI/CD pipeline reduces human error in deployments, ensures every change is tested before reaching production, and enables rapid rollback if issues occur.
Measurable Target / Threshold	GitHub Actions pipeline: triggered on every pull request to main branch. Pipeline stages: (1) Static code analysis (ESLint, TypeScript checks), (2) Unit test suite, (3) Integration test suite, (4) Security scan (OWASP ZAP, dependency audit), (5) Accessibility scan (axe-core), (6) Docker image build, (7) Push to container registry, (8) Deploy to Staging environment, (9) Health check — automatic rollback if health check fails, (10) Manual approval gate before promotion to Production. Pipeline execution time target: ≤ 15 minutes for full suite. Zero manual steps between code merge and Staging deployment.
Test / Verification Method	Review of CI/CD pipeline configuration in GitHub Actions. Test: introduce a failing unit test — verify pipeline blocks deployment. Review of deployment logs for evidence of health-checked rollouts.
Risk if Not Met	Untested code reaches production; manual deployment errors; slow release cycles; inability to roll back safely.
Source	Revised Inception Report v1.2 Section 4.1.8; Inception Report v3.0

11. Compliance and Standards Summary

The following table summarises all regulatory standards, frameworks, and best practices that the Bridge to Bangladesh platform must comply with at production go-live.

Standard / Framework	Applicability	Compliance Requirement
UK GDPR (Data Protection Act 2018)	All user data — UK-based diaspora and UK operations	Full compliance — non-negotiable. Confirmed by Oxfam GDPR focal point.
EU GDPR	EU-resident users (secondary applicability)	Aligned to UK GDPR — same controls applied throughout.
OWASP Top 10	Web application security	Zero critical/high vulnerabilities at go-live. Verified by penetration test.
WCAG 2.1 Level AA	Web application accessibility	Full compliance required — mandated by ToR Section 3.2.
ISO 27001 (aligned)	Information security management	Controls aligned to ISO 27001 — formal certification not required by ToR but alignment mandatory.
PCI DSS (via gateway)	Payment card data — Donation Module	Handled by PCI-compliant payment gateway (Stripe or equivalent). No raw card data stored on platform.
UK PECR	Cookie consent — web application	Compliant cookie consent mechanism required before non-essential cookies set.
OWASP Mobile Top 10	Flutter mobile application	Mobile application security review against OWASP Mobile Top 10 before beta submission.
Oxfam Information Security Policy	All data handling and access control	Platform must be consistent with Oxfam internal information security policy — confirmed in GDPR/Security session.